

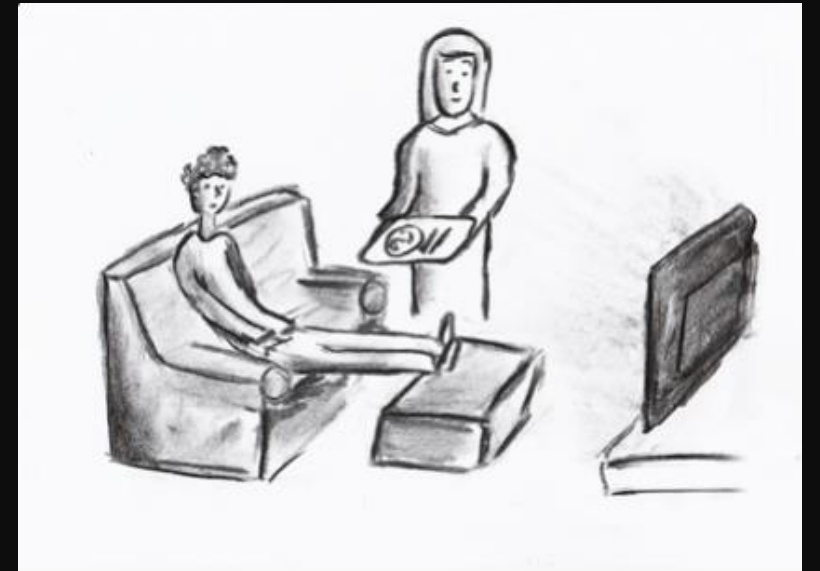
Responsible use of Large Language Models for research

Lecturers

Lodewijk Pet



Bram van Dijk



Mini quiz

Raise your hand if it is yes(ish)

1. I avoid Large Language Models (LLM)
2. I occasionally uses LLMs (anywhere)
3. I often use LLMs in personal life
4. I have experimented with LLM in work
5. I integrated LLM in work
6. My life depends on LLMs

Goals of today and contents

Gain AI literacy

1. Understand LLMs in their historical context
2. Understand how LLMs work and current developments
3. Know what is allowed and what is possible; rules and tools
4. Recognize LLM jargon and clear up confusion
5. Practice with setting up and using LLMs locally

What you can expect of RULLMR

Three similar broad and theoretical introductory meetings (9, 16, 23 sep);

- Orientation w.r.t. the topic, building basic AI literacy

Follow-up with 7 focused, practical sessions on local LLM use for research:

1. Literature search
2. Data collection
3. Study design
4. Data analysis
5. Scientific writing
6. Peer review
7. Grant writing

Course set-up

Today is the basis for all other sessions

For sessions: work in pairs (big advantage)

Bring a laptop: **Open an LLM chat right now**

Hard rule: **Don't use sensitive data**

Good practice: Keep a log of your work

- Model and model settings used
- Prompt description
- Expected output
- Evaluation

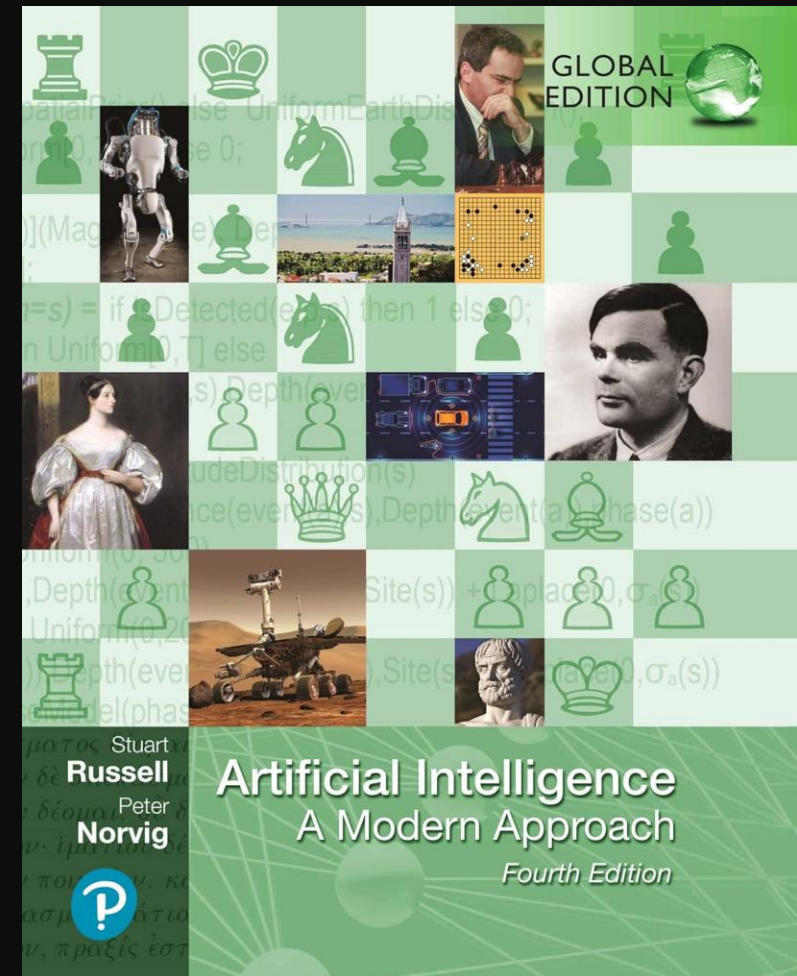
Think for yourself

What is your guiding principle?

“I am fine with someone using LLMs in research, as long as ...”

1. Place LLMs in their historical context

History of AI and LLMs



Russell & Norvig (2022)

Philosophical Foundations

- **What is intelligence?** How we *think* and *act*, i.e. perceive, understand, predict, and manipulate, a world far larger than ourselves.
 - **What is artificial intelligence?** Understanding and building entities that can compute how to think and act in a variety of situations
-

	Thinking	Acting
Human	Thinking Humanly	Acting Humanly
Rational	Thinking Rationally	Acting Rationally

Philosophical Foundations

Thinking humanly

- Express theories of the mind as computer programs so that a computer mimics human thinking
- Example: LLMs as models of cognition

Acting humanly

- Focus on behaviour and 'avoid' philosophy
- Example: Turing test

Thinking rationally

- Find the right way of thinking through logic and/or probability
- Example: Aristotle (4th century BCE), Leibniz (17th century), Boole (19th century)

Acting rationally

- A system that attempts to achieve the best possible outcome through logic and/or probability
- Example: self-driving car

Birth of 'Artificial Intelligence' as a field (1956)

- Dartmouth Workshop (1956)
 - Workshop proposal coined the term 'Artificial Intelligence'
 - Key figures: John McCarthy, Marvin Minsky, Herbert Simon, Nathaniel Rochester, Claude Shannon



Photo by Nathaniel Rochester

Birth of 'Artificial Intelligence' as a field (1956)

- *Note: Few breakthroughs came out of the meeting*

A Proposal for the
DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

The following are some aspects of the artificial intelligence problem:

1) Automatic Computers

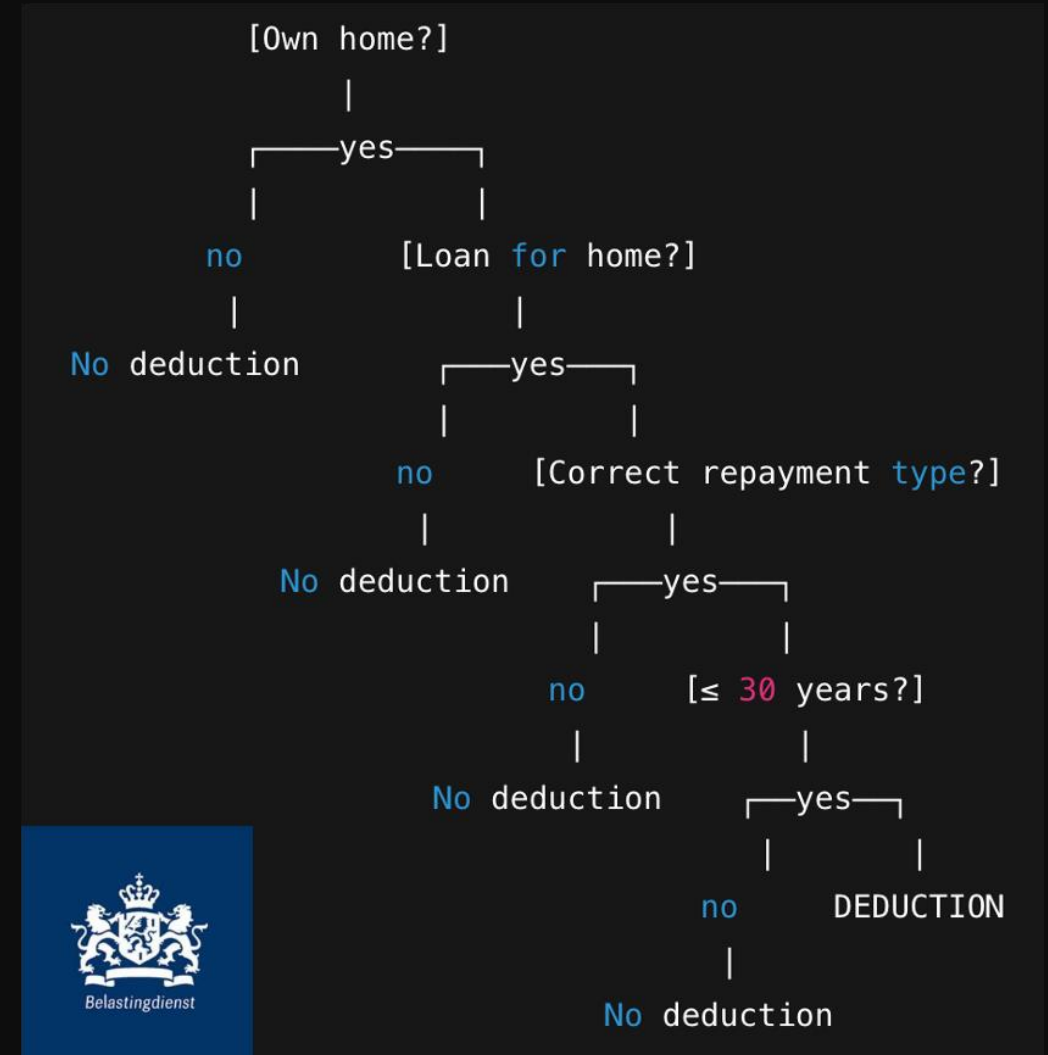
If a machine can do a job, then an automatic calculator can be programmed to simulate the machine. The speeds and

McCarthy et al. (1956)

Good Old-Fashioned AI (GOFAI) 1950s–1980s

- Core idea: *intelligence = symbols + logic*
 - Symbolic reasoning using logic-based systems (e.g. if–then rules) and symbols for representation
 - Successful in many 'microworlds', e.g. within maths and board games
 - Good: these systems were transparent, precise, deterministic
 - Bad: they generalised very poorly
- GOFAI is (still) all around us!

GOFAI example



AI winters

1970s & late

1980s

Herbert Simon in 1958:

It is not my aim to surprise or shock you—but the simplest way I can summarize is to say that there are now in the world machines that think, that learn and that create. Moreover, their ability to do these things is going to increase rapidly until—in a visible future—the range of problems they can handle will be coextensive with the range to which the human mind has been applied.

Operations Research (1958)

AI winters

1970s & late

1980s

- Why AI stalled
 - 'Overpromise, underdeliver'
 - Lack of real-world robustness
 - Decreasing funding
- Two major AI winters
 - 1970s: most real-world problems were intractable for symbolic AI
 - Late 1980s: infusing systems with 'expert knowledge' became too expensive and fragile

Probabilistic and machine learning (2006-2016)

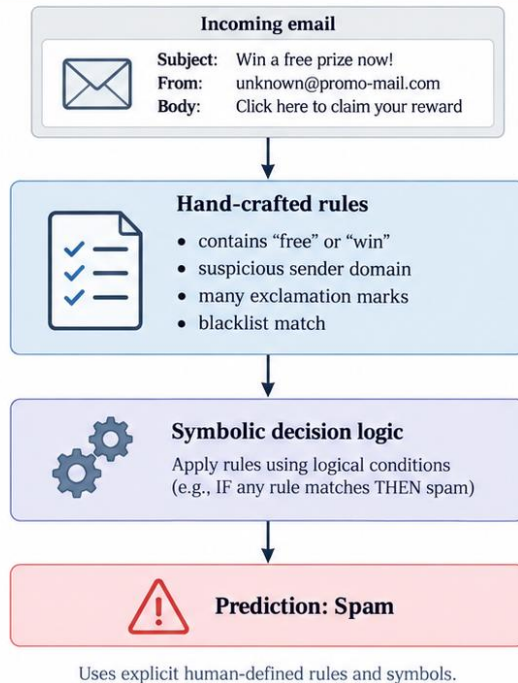
- New research directions
 - Doubts about the *deductive* approach to a messy world
 - The *connectionist* approach learned directly from real-world data successfully due to:
 - New hardware (e.g. GPUs)
 - Data availability (growth of the web)
 - Example: spam classification

Probabilistic and machine learning (2006-2016)

the	a	win	free	you	money	won	prize	!	...
0	1	1	1	0	0	0	1	1	...

Spam Classification: GOFAI vs. Connectionist Model

GOFAI (Rule-Based / Symbolic AI)

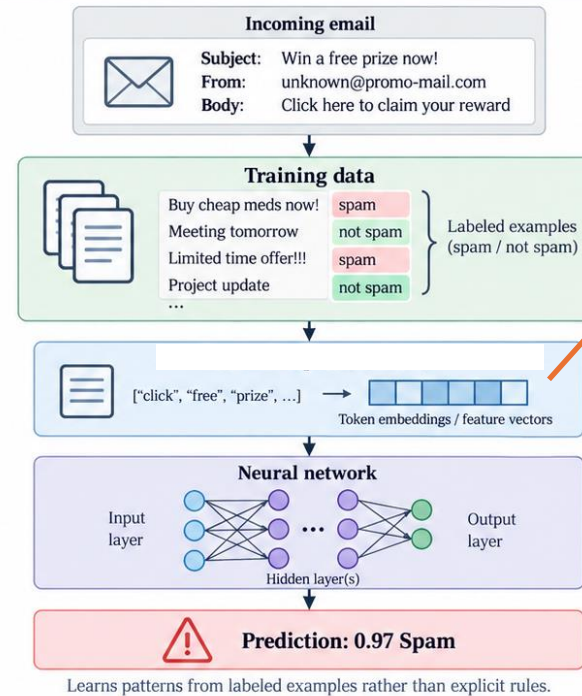


Strengths:
interpretable, easy to inspect



Limitations:
brittle, misses new patterns,
rule maintenance needed

Connectionist Model (Neural Network)



Strengths:
adapts to complex patterns,
can generalize



Limitations:
less transparent, needs data
and training

Probabilistic and machine learning (2006-2016)

- New research directions
 - Breakthroughs in speech transcription, object recognition, medical diagnosis, game play, machine translation, and...
 - Language generation!

2. Understand how LLMs work

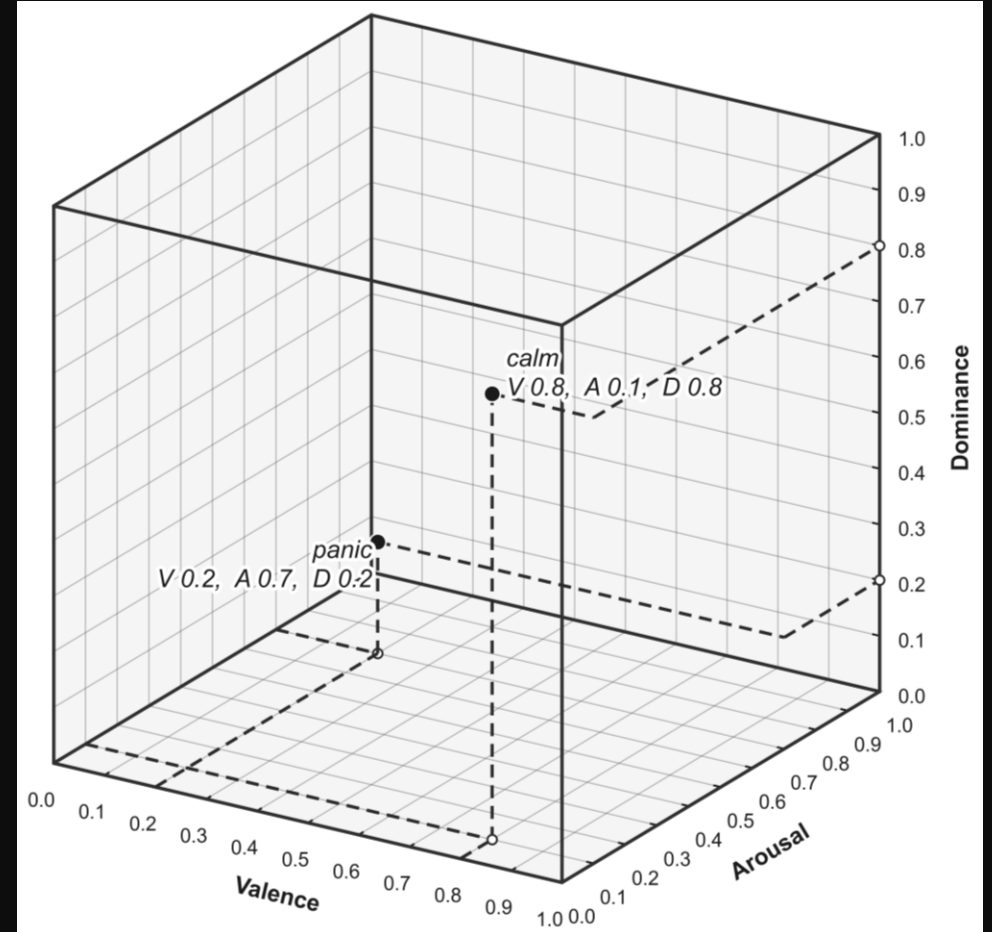
Meaning representation, training, inference, and current developments

LLM mechanics: representing meaning

- Words can be represented with numbers (parameters/weights) in a vector space
 - Osgood et al.'s (1957) point representation of word meaning was one of the first examples, closely related to the VAD model of word meaning by Mehrabian & Russel (1974) >>>
- We can 'train' our parameters/weights by asking many people to rate words, and average ratings

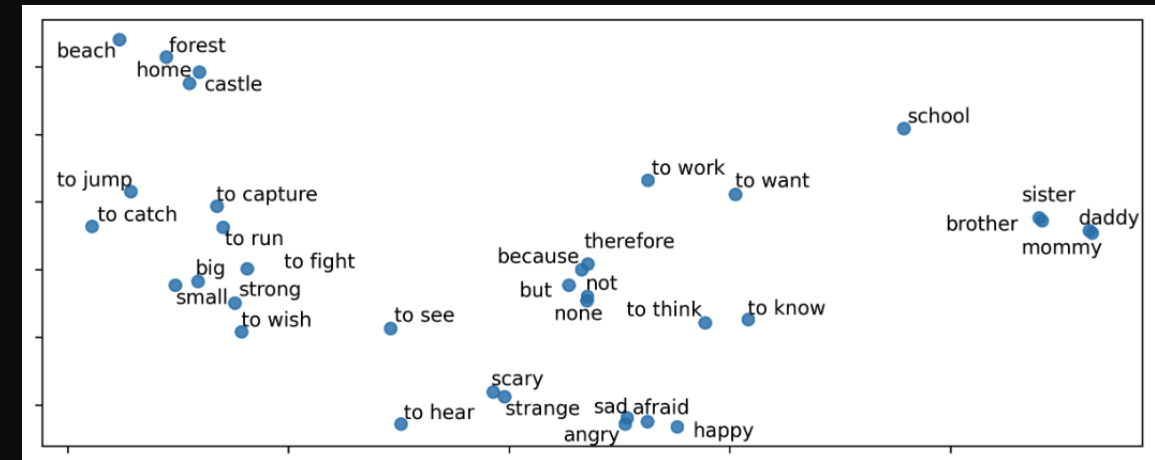


Osgood et al. (1957). *The Measurement of Meaning*.
Mehrabian & Russel. (1974). *Distinguishing anger and anxiety in terms of emotional response factors*.



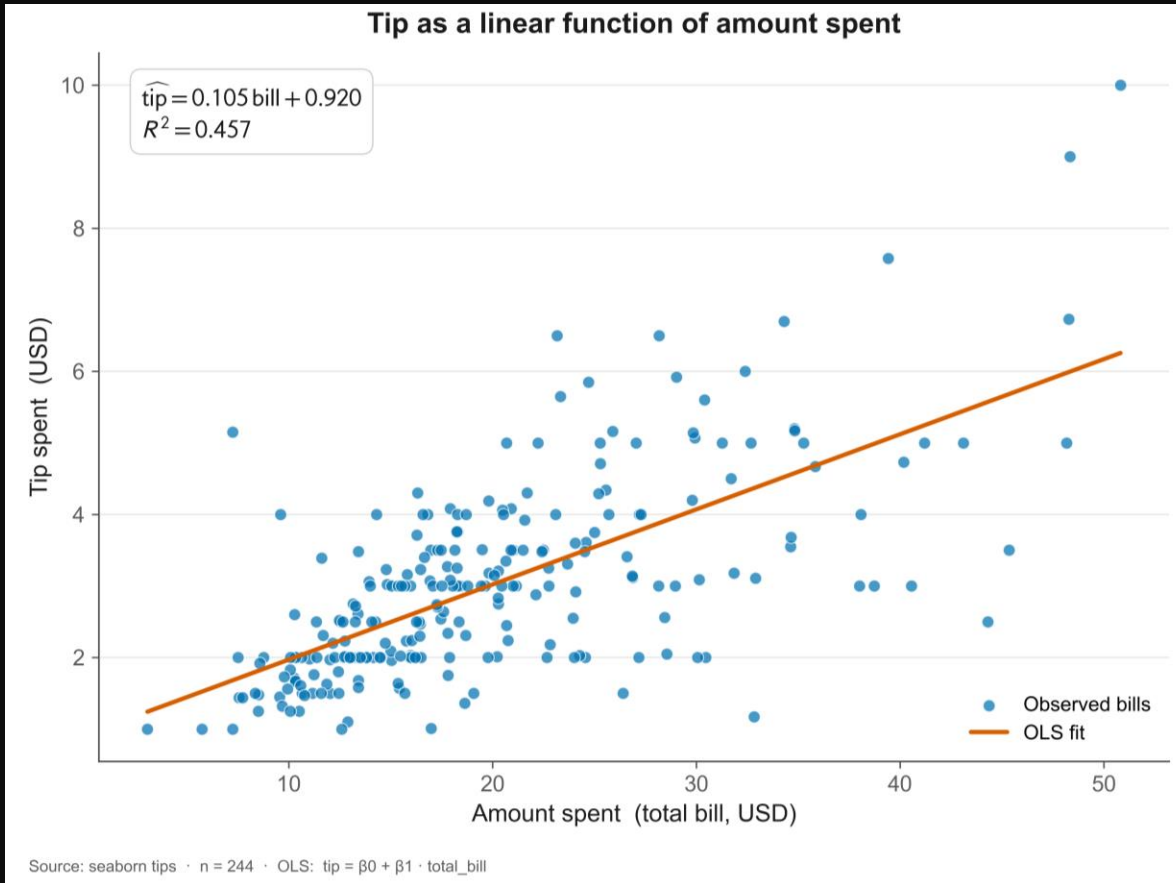
LLM mechanics: representing meaning

- Before the LLM era, the field focused on automatically learning representations of words, with words represented as vectors in a high-dimensional space (in the plot compressed to just two)
- We need a prediction task to train our parameters/weights



Van Dijk et al. (2023). *ChiSCor: a Corpus of Freely-told Fantasy Stories by Dutch Children for Computational Linguistics and Cognitive Science*

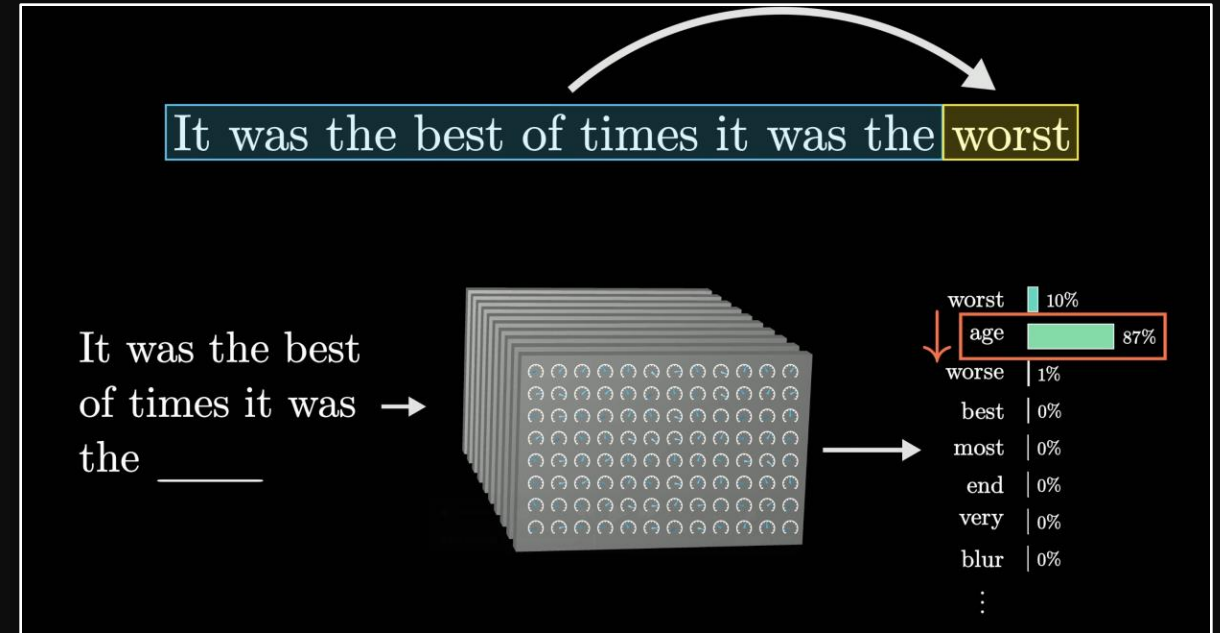
LLM mechanics: predicting tip



- A familiar case
 - Predicting y from x using some learnable parameters, some dataset, and some algorithm
 - For a prediction, we multiply inputs with weights and add bias

LLM mechanics: pre-training

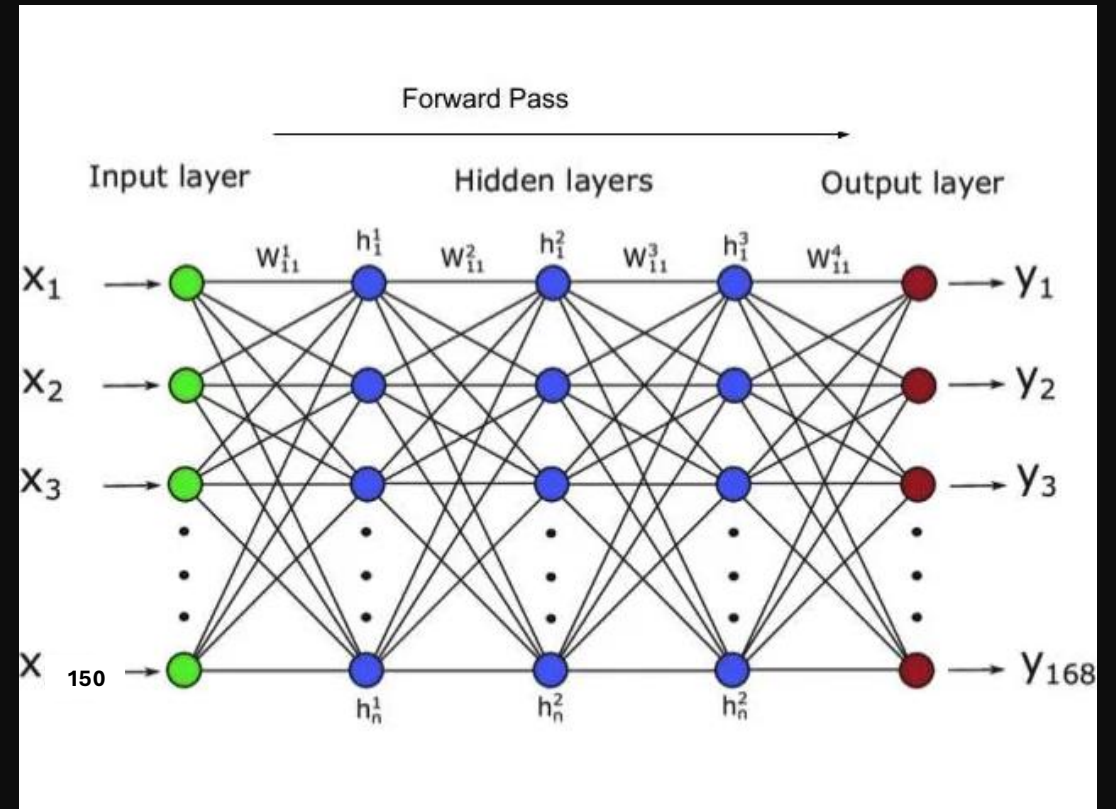
- Again a prediction task, but slightly different:
 - Features are context words (vectors),
 - Outputs are probabilities over an entire vocabulary,
 - Model is a big neural network (complex, many parameters)



threeblueonebrown.com

LLM mechanics: pre-training

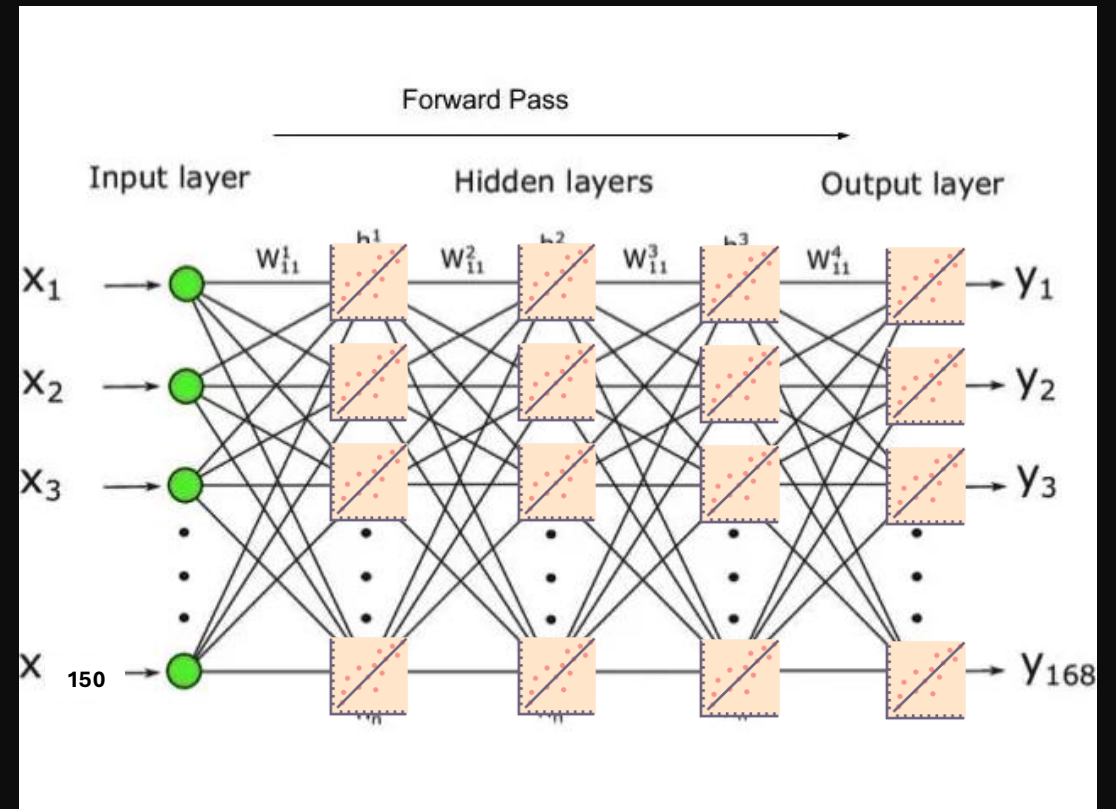
- You can think of a simple neural network as:
 - a 'stack' of linear regressions with non-linear activations (like logistic regression with the sigmoid function)



Caspar Bannink (medium.com)

LLM mechanics: pre-training

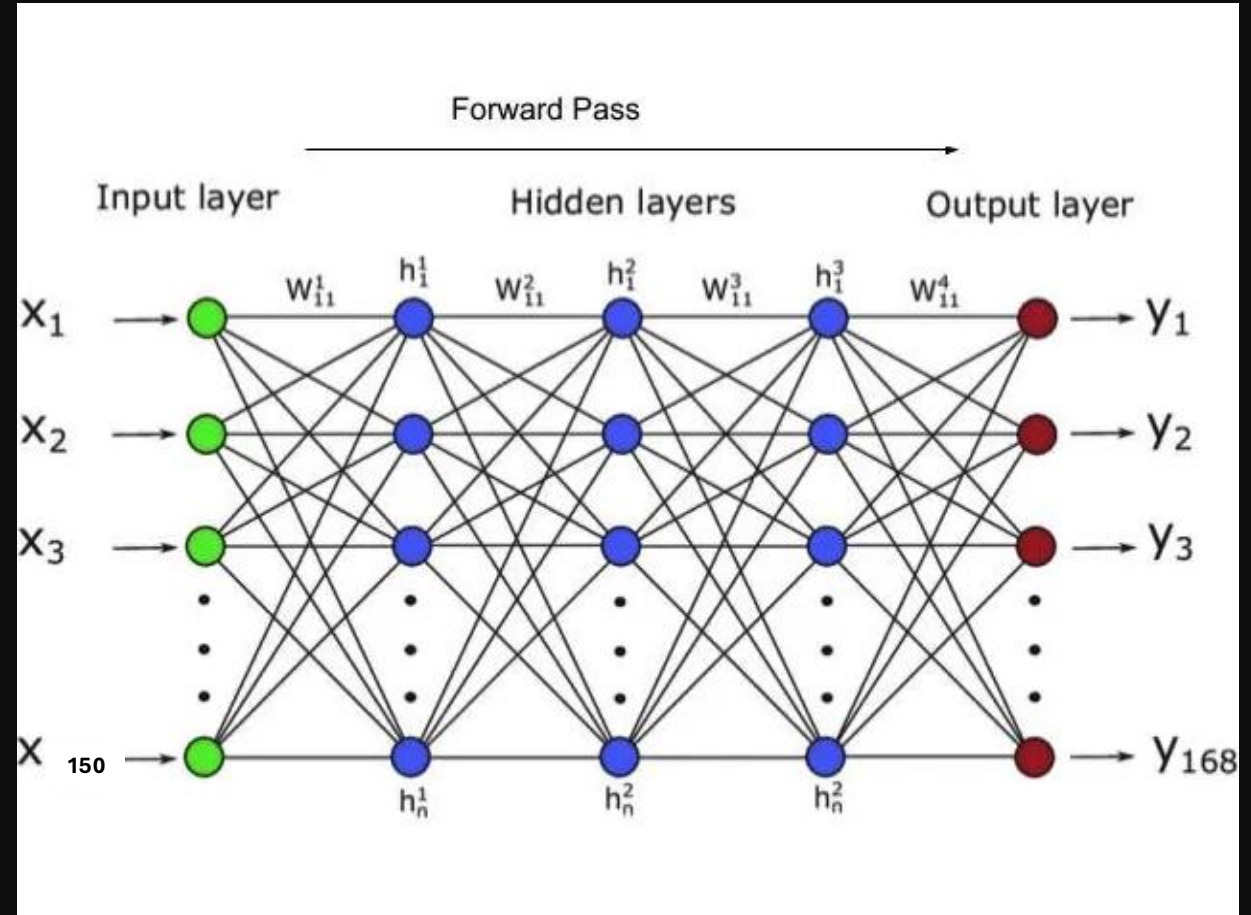
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LLM mechanics: pre-training

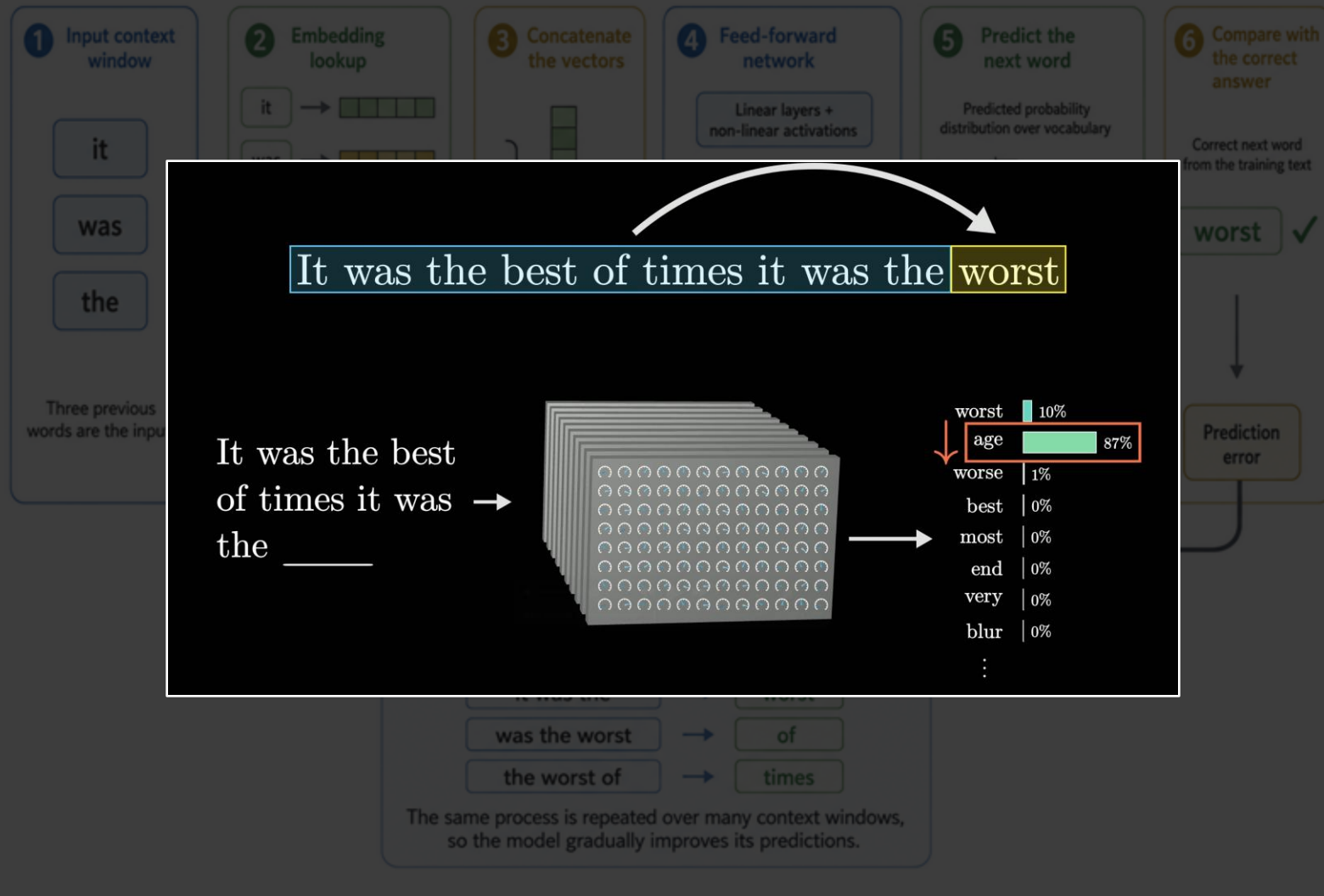
- You can think of a simple neural network as:
 - a 'stack' of linear regressions
 - Where at each neuron, inputs are multiplied by weights and a bias is added
 - Neuron output is pushed through a non-linear activation and input to the next layer
 - Last layer typically adapted to task



Caspar Bannink (medium.com)

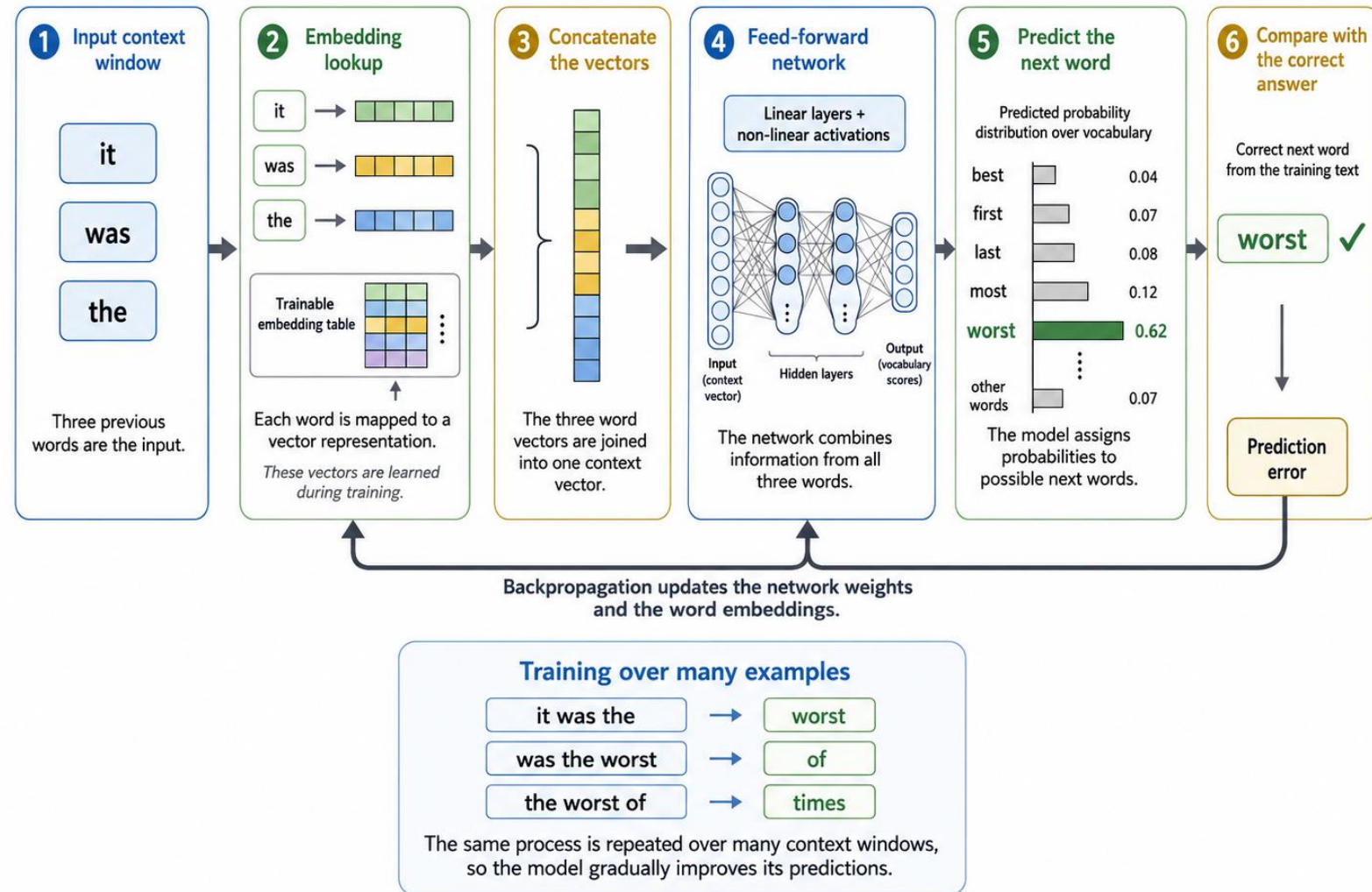
Training a Simple Feed-Forward Language Model

Example: use the context 'it was the' to predict the next word 'worst'



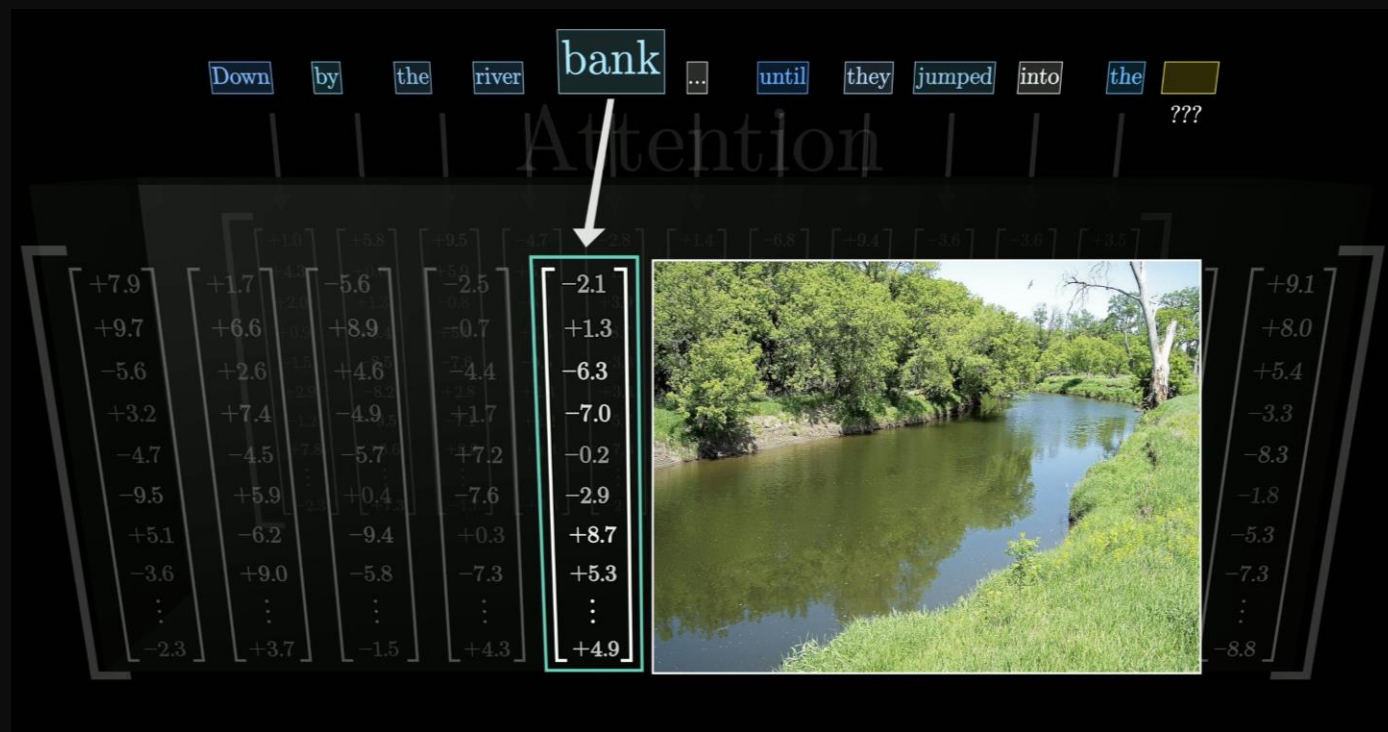
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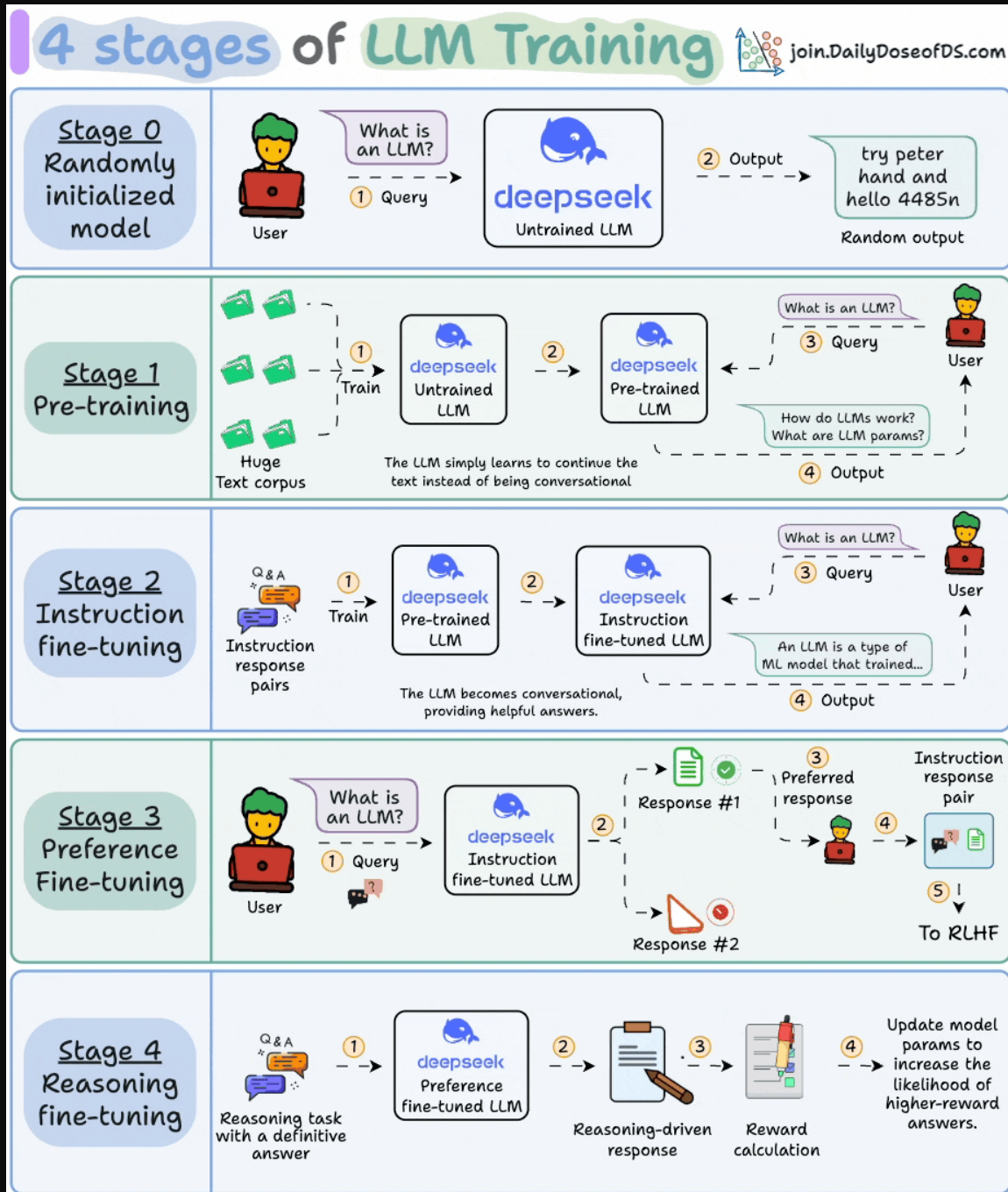
LLM mechanics: pre-training

- The neural nets underlying LLMs are more complex:
 - One important difference is the *Transformer* architecture, which allows the network to transform a word vector based on *attention* to context words
 - Transformer was introduced in the famous *Attention is all you need* paper by Vaswani et al. (2017)



LLM mechanics: post-training

- What you need to know next time uncle Ben at a birthday party reduces LLM training to just next-word prediction: the story is much more complex! >>>



LLM mechanics: post-training

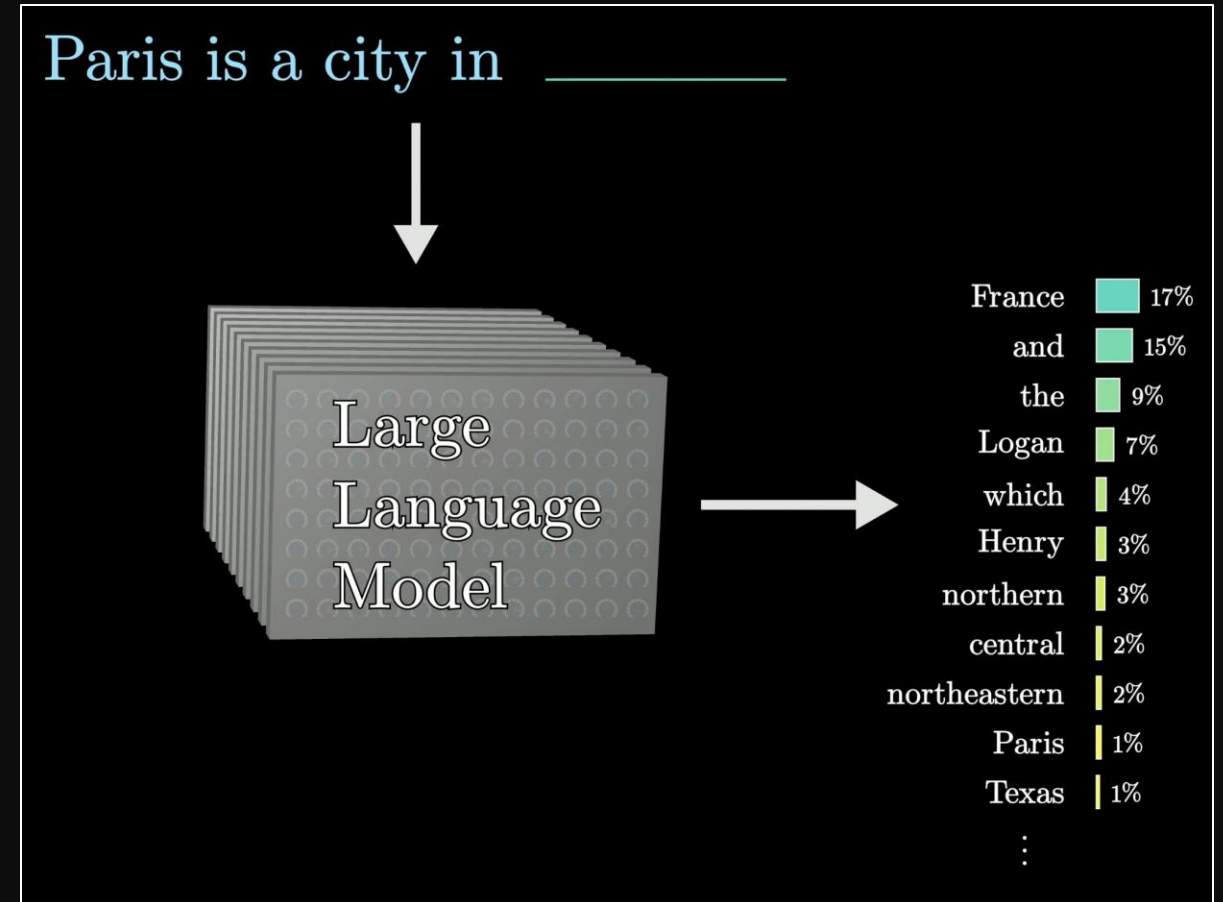
- This learning from human feedback is sometimes done out of sight



<https://www.404media.co/inside-project-lily-the-humans-reading-your-chatgpt-chats/>

LLM mechanics: natural language...

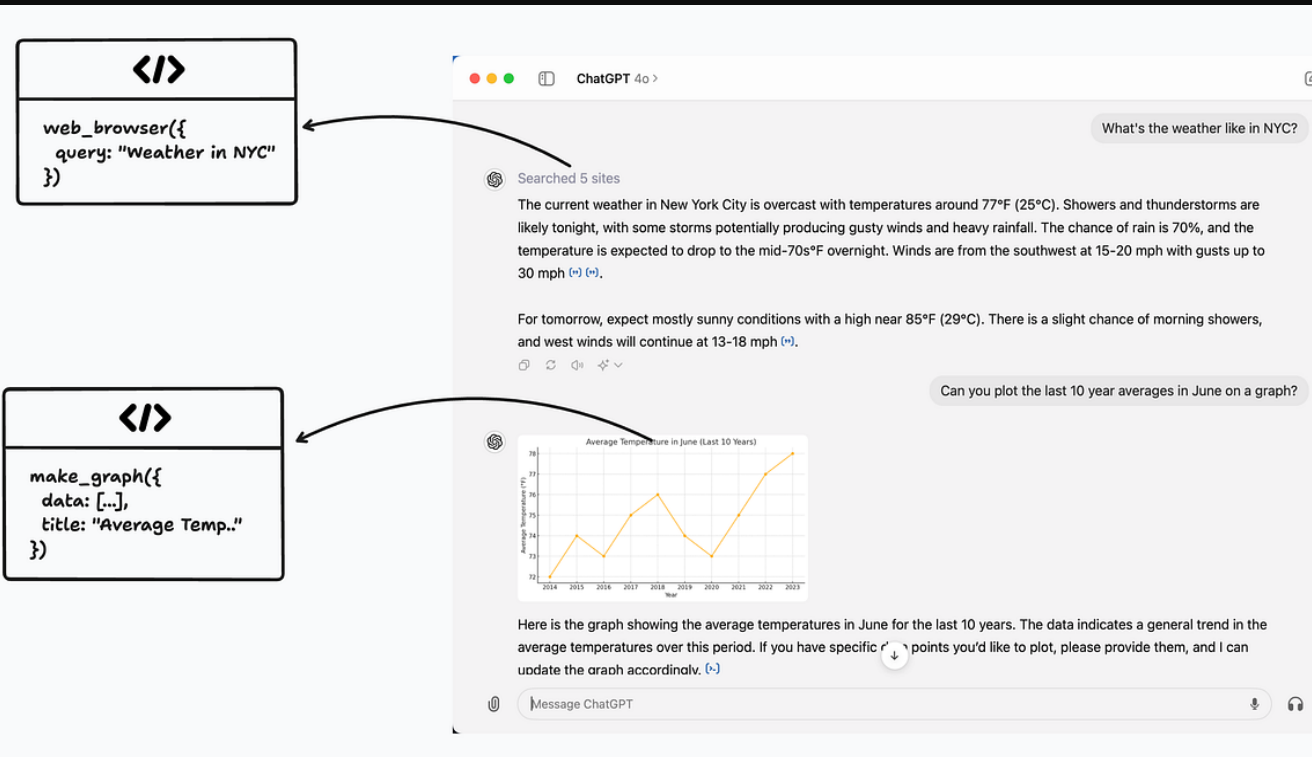
- After training we end up with a model that we use for *inference*. Its output can be *natural language*...



threeblueonebrown.com

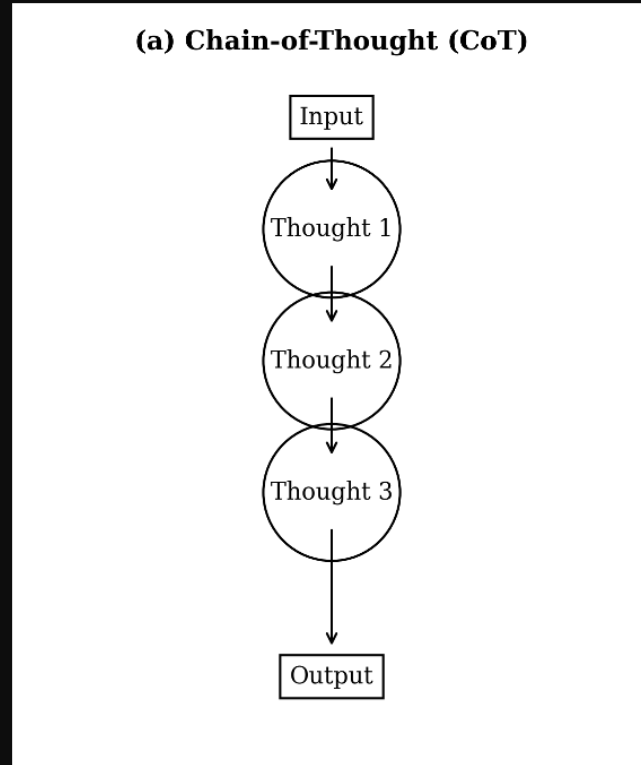
LLM mechanics: ...tool calls...

- But it can also be a *tool call*, which makes the LLM *agentic*:
 - with a tool call the LLM can initiate tasks like web search, fetching mails, generating plots, etc.
 - The LLM is still predicting next words, but in a particular format. Another program continuously inspects LLM output for tool calls, executes them, and feeds the results back to the LLM
 - LLMs learn to associate certain inputs with tool calls, e.g. because the request requires current information
 - This makes LLMs less error prone and more robust
 - Chatbots are 2025, 2026 is the year of agents ;)



Jared Zoneraich (medium.com)

LLM mechanics: ...and reasoning



- In LLM post-training on complex problems, reasoning traces emerge because they help finding the correct answer
 - 'The user asks X, let's break it down'
 - 'Assume P, this implies Q'
 - Revisiting earlier claims ("wait, I must check P")
 - Reasoning traces do not necessarily mirror model internals
 - LLMs typically spend more tokens on 'complex' inputs, i.e. reasoning is about spending compute
-

3. What is allowed? Rules and tools

Laws, regulations, and norms
Providers & models

Three layers

1. The law
2. Agreements
3. Norms

The restrictions: Law

ALL RESEARCH & DATA

1. GDPR
2. EHDS
3. Dutch Civil Code
Wet BIG
Dutch Criminal Code
4. Research exemptions: UAVG
5. Copyright
6. Database rights
7. Trade secrets

AI-ACT

1. Does the AI Act apply?
2. Prohibited use?
3. High-risk AI use?

EXTERNAL PARTY INVOLVED: GDPR

1. Can I send these data?
2. Data disclosed to an unauthorised party?

The restrictions: Law

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The restrictions: Law

	Local statistics (R/SPSS/Python)	Fully local ChatGPT-like model	External ChatGPT/Claude
GDPR on research data	✓	✓	✓
Health-data rules	✓	✓	✓
Security requirements	✓	✓	✓
Copyright / trade secrets	✓	✓	✓
External processor issue	—	—	✓
International transfer issue	—	—	✓
AI Act	potentially	✓	✓
Automatically high-risk	No	No	No

The restrictions: Agreements and institutional rules

Your employer

IT/security policy, approved software,
Data-classification rules,
AI policy

Funding / data providers / publishers

Grant conditions
Data-access agreements
Database licences
Publication embargoes

Ethical / nWMO agreements

Informed consent
Approved study protocol
Data-management plan

External parties

Non Disclosure Agreement
Data Use Agreement
Data Processing Agreement
Collaboration/consortium agreement
Material Transfer Agreement

Governance in the LUMC

LUMC policy in brief:

- Use closed source LLMs only with publicly available data
- Use of 'internal' (but not sensitive) data is possible with providers with which an agreement was signed (currently only Copilot in your M365 environment)
- New policy based on changed context and requirements in research is work in progress

See the LUMC policy on use of AI at <https://lumc.zenya.work/QC/47-PXM-4>



Governance in the LUMC

- Many of you probably use (paid or unpaid) LLMs for coding, proofreading, reviewing, polishing powerpoints, etc.
 - We **cannot** provide hard answers on what is and what isn't allowed!
 - Try to use common sense: if you write code with a LLM and the repository will be open access, or if you use a LLM to help explaining a hard part of a paper to us don't seem like problematic issues
 - In the end you as LLM user, for whatever task, are always responsible
 - Copilot likely has poor integration with your local workspace but as chatbot it has improved a lot
-



The restrictions:

Community norms

Research integrity codes

European Code of Conduct for
Research integrity

Journal & publisher policies

Authorship rules

Peer-review confidentiality

AI disclosure

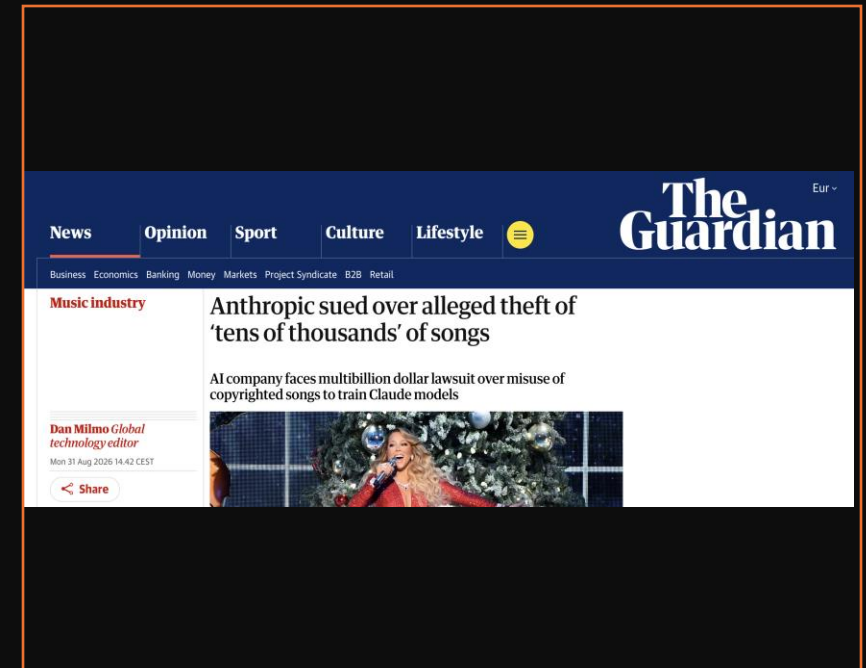
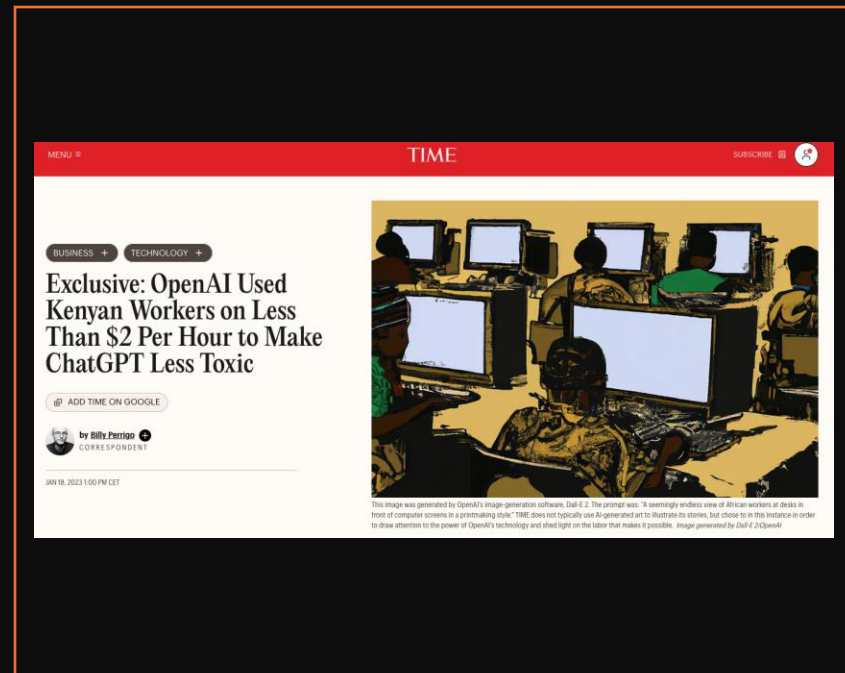
Image generation / alteration

Ethics

Consider how the data for the models is collected.

Follow the news.

Don't be fooled by nice messages.



Sustainability

How much energy does my prompt use?

Depends on many factors

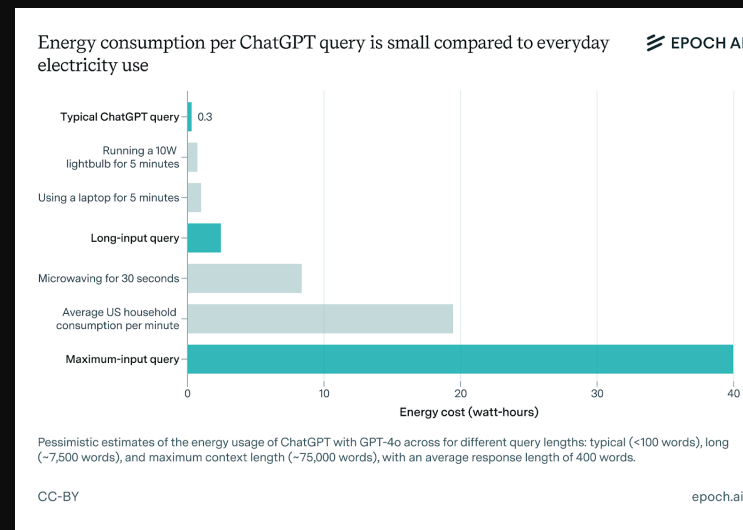
- Prompt & output length
- Model and settings used
- GPU type, location of data center
- ...

But, less than you might think.

But... industry-reported energy consumption

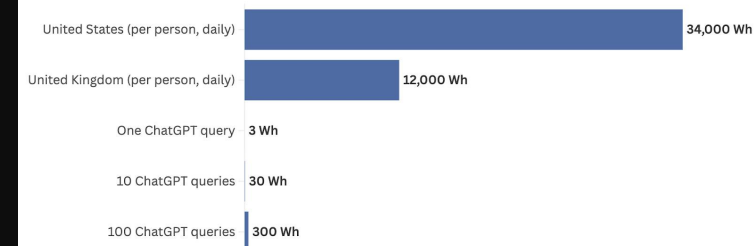
However... Reduced energy use is corporate interest

Note: this leaves out costs of LLM training



How does the electricity use of ChatGPT compare to the daily average footprint in the UK and the US?

Measured in watt-hours. Electricity consumption for the US and UK includes household use plus services and industry within the economy.



Source: Ember Energy, Alex de Vries, and Andy Massley • This assumes that one ChatGPT query uses 3 watt-hours of electricity. More recent data suggests this could now be as low as 0.3 Wh.

Sum

How n

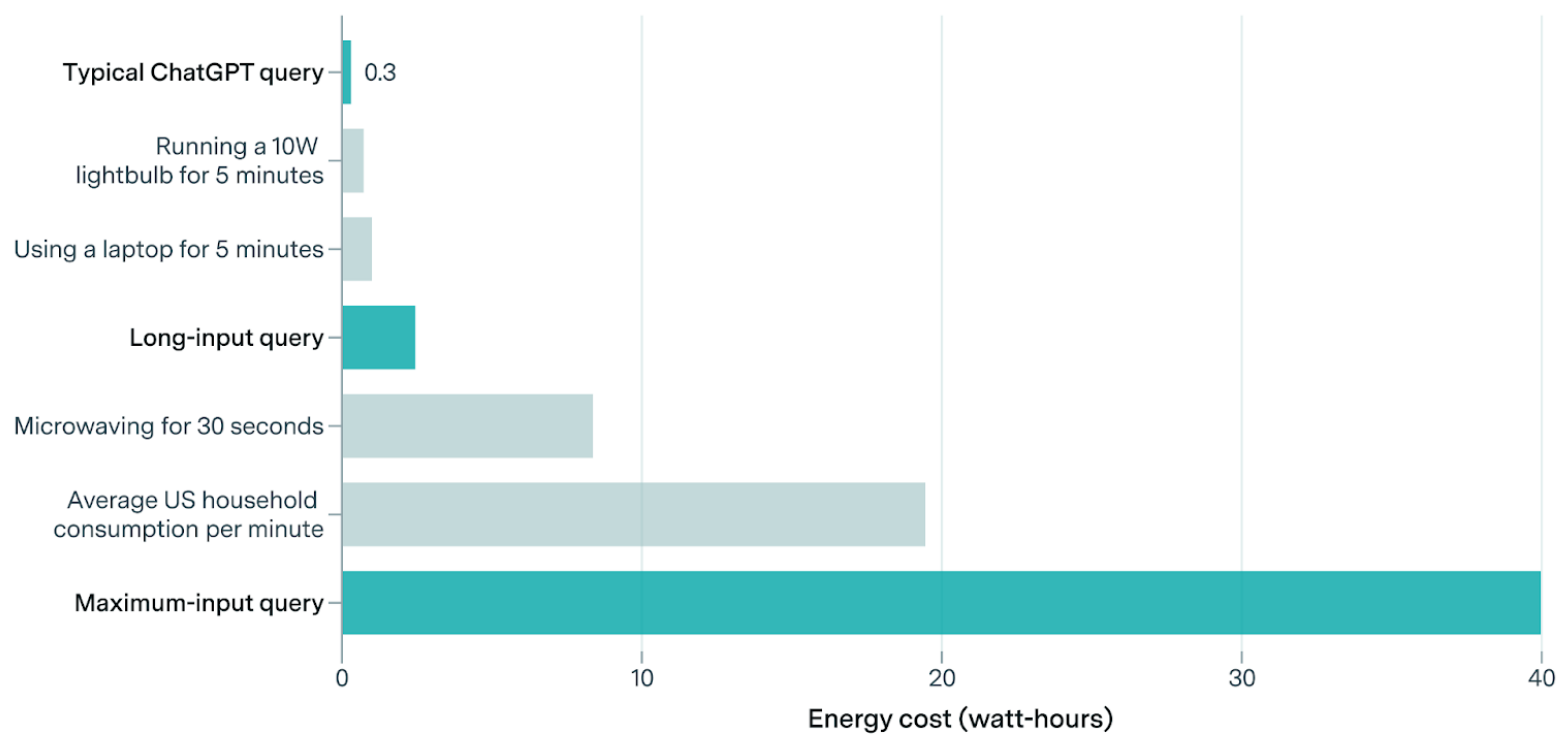
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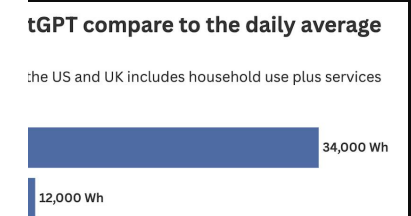
Energy consumption per ChatGPT query is small compared to everyday electricity use



Pessimistic estimates of the energy usage of ChatGPT with GPT-4o across for different query lengths: typical (<100 words), long (~7,500 words), and maximum context length (~75,000 words), with an average response length of 400 words.

CC-BY

epoch.ai

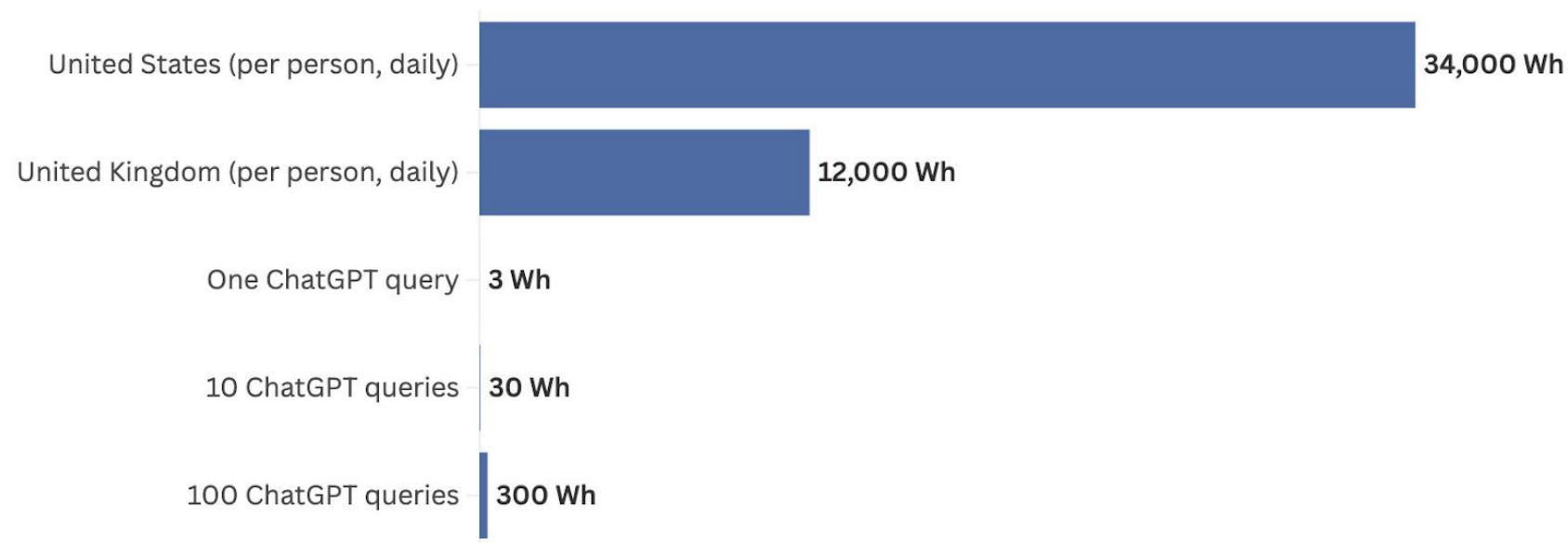


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Su

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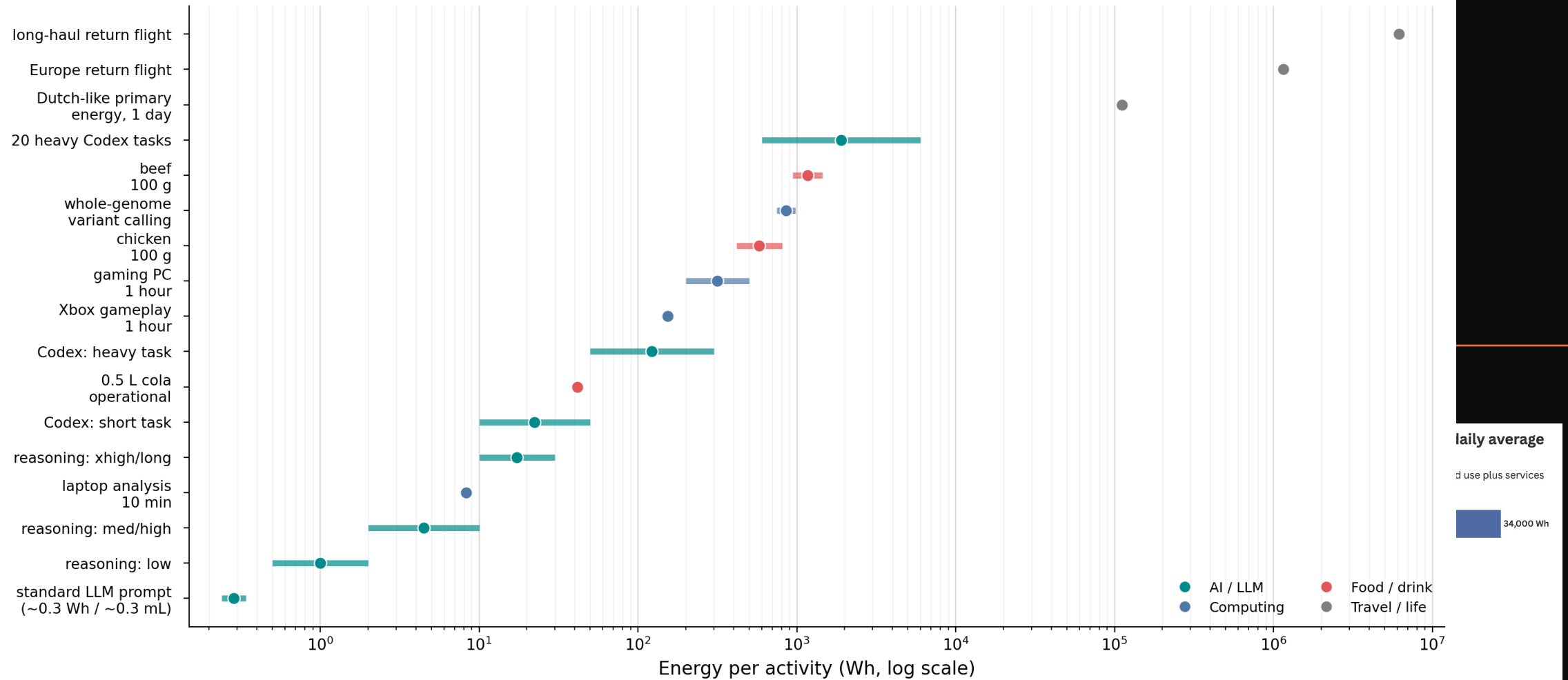
EPOCH AI

Typical ChatGPT query 0.3

Energy consumption per ChatGPT query is small compared to everyday electricity use

EPOCH AI

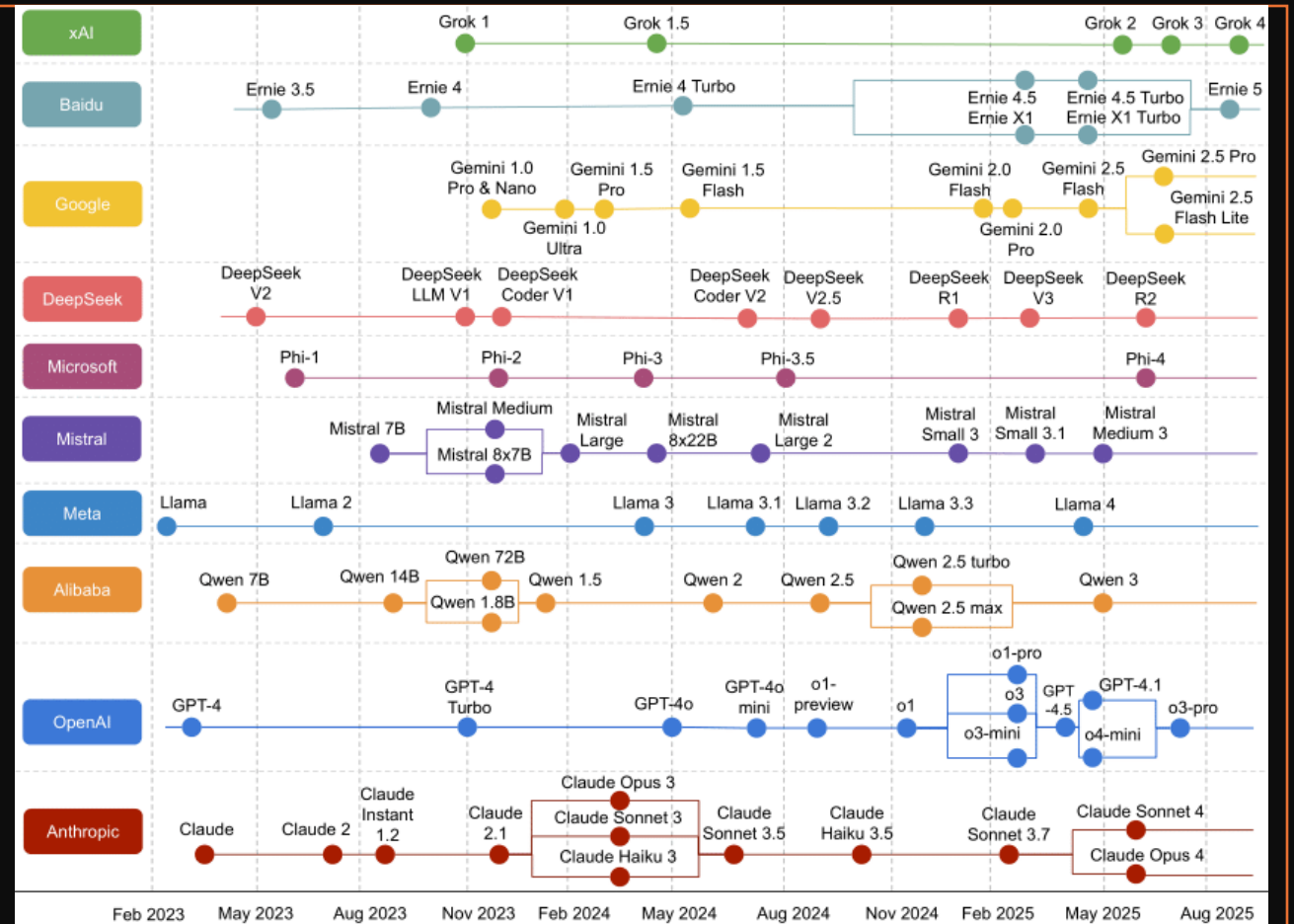
Energy use: AI prompts and agents in context



Lines show min-max ranges; dots show geometric midpoints. Reasoning/Codex rows are scenario estimates, not official GPT energy data.

~3 watt-hours of

Providers and Models



Le & Abel (2025). How Well Do LLMs Predict Prerequisite Skills? Zero-Shot Comparison to Expert-Defined Concepts

Providers

OpenAI and Anthropic: create applications

- Multiple AI systems, Code integration, Web access
- “Reasoning”, “Deep research”
- Agentic frameworks

Carefully constructed packages

Combination of tools

Model and benchmark abundance

MODEL	OVERALL ▼	REASONING	CODING	AGENTIC CODING	MATHEMATICS	DATA ANALYSIS	LANGUAGE	INSTRUCTION FOLLOWING	COST PER SUCCESSFUL TASK
▸ Claude Fable 5.1 Max Effort	83.4	91.7	86.4	66.1	97.0	80.3	89.5	73.0	\$1.212
▸ Claude Fable 5 Max Effort	83.0	89.7	86.0	62.2	96.0	80.5	90.7	75.8	\$1.439
▸ GPT-5.6 Sol Max Effort	81.0	91.7	83.9	56.2	96.2	79.8	87.7	71.8	\$0.515
▸ GPT-5.5 Thinking xHigh Effort	80.2	89.7	82.1	54.0	95.9	81.6	87.4	70.7	\$0.435
▸ Claude 5 Opus Thinking Max Effort	80.1	91.2	81.4	65.2	95.7	74.6	88.7	63.8	\$0.699
▸ Kimi K3 open	79.2	90.7	81.4	62.2	84.4	78.7	85.5	71.4	\$0.348
▸ Gemini 3.7 Flash High	78.8	87.8	78.9	58.3	93.5	68.0	85.5	79.9	\$0.157
▸ Qwen 3.8 Max open	78.5	88.2	72.9	64.6	91.3	78.4	79.7	74.1	\$0.275
▸ Grok 4.6	78.0	90.5	76.8	57.0	92.6	73.9	83.7	71.9	\$0.207

1 Sept. 2026 overview of livebench.AI

Model and benchmark abundance

RANK	MODEL	LLM STATS  	REASONING 	CODING 	AGENT 	CODE ARENA 	PARAMS 	CONTEXT 	SPEED 	PRICING \$/M 
1	 GPT-5.6 Sol - OpenAI	56.6	55.9	48.9	42.6	2,134	—	1.1M	105 c/s	\$7.78
2	 Claude Mythos Preview - UNRELEASED Anthropic	55.6	56.3	45.3	36.8	—	—	—	—	—
3	 Claude Opus 5 ↓ 1 Anthropic	55.3	54.3	41.3	39.5	2,668	—	1.0M	87 c/s	\$7.22
4	 Claude Fable 5 ↓ 1 Anthropic	55.1	52.4	47.2	40.9	2,003	—	1.0M	103 c/s	\$14.44
5	 Claude Fable 5.1 - NEW Anthropic	54.8	52.1	—	39.3	—	—	1.0M	—	\$14.44
6	 Kimi K3 ↓ 2 Moonshot AI	54.5	53.3	45.0	40.3	1,816	2.8T	1.0M	87 c/s	\$4.33
7	 GLM-5.3 ↓ 2 Zhipu AI	54.1	53.2	44.5	40.2	—	753B	1.0M	77 c/s	\$1.73
8	 DeepSeek-V4-Pro-0813 ↓ 2 DeepSeek	52.9	50.3	43.0	39.0	—	1.6T	1.0M	120 c/s	\$0.48

1 Sept. 2026 overview of llm-stats.com

LLM providers...

OpenAI, Anthropic, Meta, Alibaba, Moonshot, Mistral, ...

Are friends, enemies, competitors:

- They help
- Compete
- (Political) enemies

Do not underestimate their influence and power

Your data is their gold

They compete

This applies to providers from both open and closed source LLMs

BBC

China's Moonshot AI stole from Anthropic, Trump tech adviser says

23 July 2026

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Add as preferred on Google

Osmond Chia
Business reporter



Kimi K3 is an AI model developed by Beijing-based Moonshot AI

A White House adviser has accused China's Moonshot AI of a "large scale" effort to steal the capabilities of top US artificial intelligence (AI) models.

US President Donald Trump's Science and Technology adviser Michael Kratsios said Moonshot AI carried out the campaign through what is known as distillation - when a weaker AI model extracts answers from a stronger one.

They crave data

NIEUWS

AI-bedrijven bestoken antiquariaten met verzoeken om tweedehands boeken

Nederlandse antiquariaten worden benaderd door buitenlandse bedrijven die hun tweedehands boeken willen gebruiken voor het voeden van kunstmatige intelligentie. Dat meldt nieuwszender BNR.

Volkskrant, June 25 2026

Closed-source LLMs

What most people think of

- Examples: Gemini, ChatGPT, Claude, Grok
- Can handle many things: writing, coding, creating, ...
- Strength: general-purpose
- Limitation: expensive (size, computing power, ...)
- Remote access: limited control, can change

Open-weights LLMs

Less familiar

- Examples: DeepSeek, GLM, Qwen, Kimi, Llama, GPT-OSS
- Often come in different sizes
- Downloadable → Locally usable and fine-tunable
- Cheaper to run, more permissive use
- Still not full open source
 - e.g. training data and process often undisclosed

SLMs vs. LLMs

Examples: Qwen2.5, Llama 3, Mistra, Gemma

Larger is not always better

SLMs are intentionally smaller:

- ~3 to 32 billion parameters
 - Can be fine-tuned for a specific task or domain
 - Fast & controllable
 - Strengths: Focus, privacy, local, reproducible
 - Limitation: non-general, lower quality
-

What should I use?

Both provider and SLMs

Provider models are more capable in expanding your knowledge

- Powerful
- Create working environment
- Set up a SLM

Use LLMs when possible

SLMs when needed

- Costs, privacy, hardware constraints

Break – 5 min

In the mean time:

Prepare for the practical of today:

- Ask ChatGPT / Codex / Copilot how to download “Ollama”
 - Download Ollama
-

4. Recognize and clarify LLM confusion and jargon

AI capabilities

Artificial Narrow Intelligence (ANI)

- The kind we have today
- Has trouble generalising beyond training domain
- Outperforms humans in some domains
- Example: ChatGPT

Artificial General Intelligence (AGI)

- Hypothetical
- Broad human-level performance across many cognitive tasks
- Flexible learning and transfer across domains

Artificial Superintelligence (ASI)

- Speculative
- Exceeds human intelligence in many domains
- Often discussed in ethics and risk literature

AI capabilities

AI (artificial intelligence)

OpenAI hails 'new era of artificial general intelligence' with Astra model release

Astra's arrival comes days after Sam Altman, OpenAI's chief executive, described AGI as an 'irrelevant marketing term'

Robert Booth *UK technology editor*

Thu 3 Sep 2026 20.24 CEST

 Share

 Prefer the Guardian on Google



Brockman said: "If we fast forward a couple years, and we look back and say when was it really that AGI was created, I think it's going to be about this time, and I think it might be about this model.

"I think that we're at a point now where these models are solving unsolved 100-year-old math problems, but you can also use them in order to accelerate the economy [and] in order to benefit you and your personal life ... I think it's not unreasonable to feel that we are now in the AGI era."

Disentangle flavours of LLMs

- LLMs
- Reasoning LLMs
- Agentic LLMs
- LLMs as Agents
- Vision models
- Audio / speech models
- Multimodal models

Human vs. AI learning

- Humans, also learn partly by imitation and prediction. LLMs are retrievers and imitators
- Yet human learning is embedded, social, and active
- Learning is grounded in eye contact, tone of voice, and physical presence, i.e. learning is rich

Bias in LLMs

Training data reflects the language and culture that is most frequent, i.e. model inherits *bias*

- Whose voices, whose topics?
-



5. What is context?

- Everything the model can use to condition its next-word prediction, e.g.:
 - Hidden instructions (by the provider)
 - Chat history (by the user)
 - User input
 - Included documents
 - Previous tool call results, tool descriptions
 - Reasoning traces (?)
- Note that model weights are not context, but still influence the prediction
- Context rot: when information in (long) contexts loses its influence on the prediction

6. What are hallucinations?

- LLM output that does not match with facts (e.g. in our world, in our input)
 - For the LLM internally this is no different from producing 'true' output;
 - Same next token prediction task
 - Same fluency and confidence
 - Interesting parallel with human memory (more reconstructive than we typically admit)
-

5. Getting practical

Learn how to download and use local LLMs

Can I use local
models on my poor
slow LUMC
computer laptop?



Yes!

PC: Ollama

Mobile: PocketPal

- Software to run AI locally
- Run LLMs without internet
- 1000s of AI models available



System > About

 Storage

477 GB

354 GB of 477 GB used

 Graphics Card

128 MB

Intel(R) UHD Graphics 770

 Installed RAM

16,0 GB

 Processor

**12th Gen Intel(R)
Core(TM) i5-12500**

3.00 GHz

What do you need?

LLMs are software

- Download and install it
- Billions of parameters in use
- Many calculations

Hardware

- Memory (storage of the model)
- “Active” Memory (RAM)
- CPU (Main computing power)
- GPU (Graphics card)

Specs of a typical LUMC device
<<<

Why a graphics card?

Graphics cards are designed for gaming

- Gaming requires many screen updates
- Many small parallel calculations
- Better to have 10 small computers
- Compared to 1 big computer

LLMs are similar

- Many small calculations
 - That's why graphics cards are needed
-

Controlling LLMs in 2025: prompt engineering

Often overhyped but fundamentally simple
Important for small models

The practice of structuring input so the model understands:

- Task, role, constraints, output format

More advanced:

- Providing examples (few-shot prompting)
- Step-by-step reasoning cues

Prompt engineering does not change the model; It changes the context.

Context is more powerful than fine-tuning and alignment

Controlling LLMs in 2026: context engineering

LLMs should rely less on memory and on tools

This is faster, cheaper, and less error-prone

Focus on LLM's *environment a.k.a. harness*

Less on LLMs themselves.

The LLMs become more orchestrators and less 'know-it-all' devices

These environments (Claude Code etc), can be downloaded.

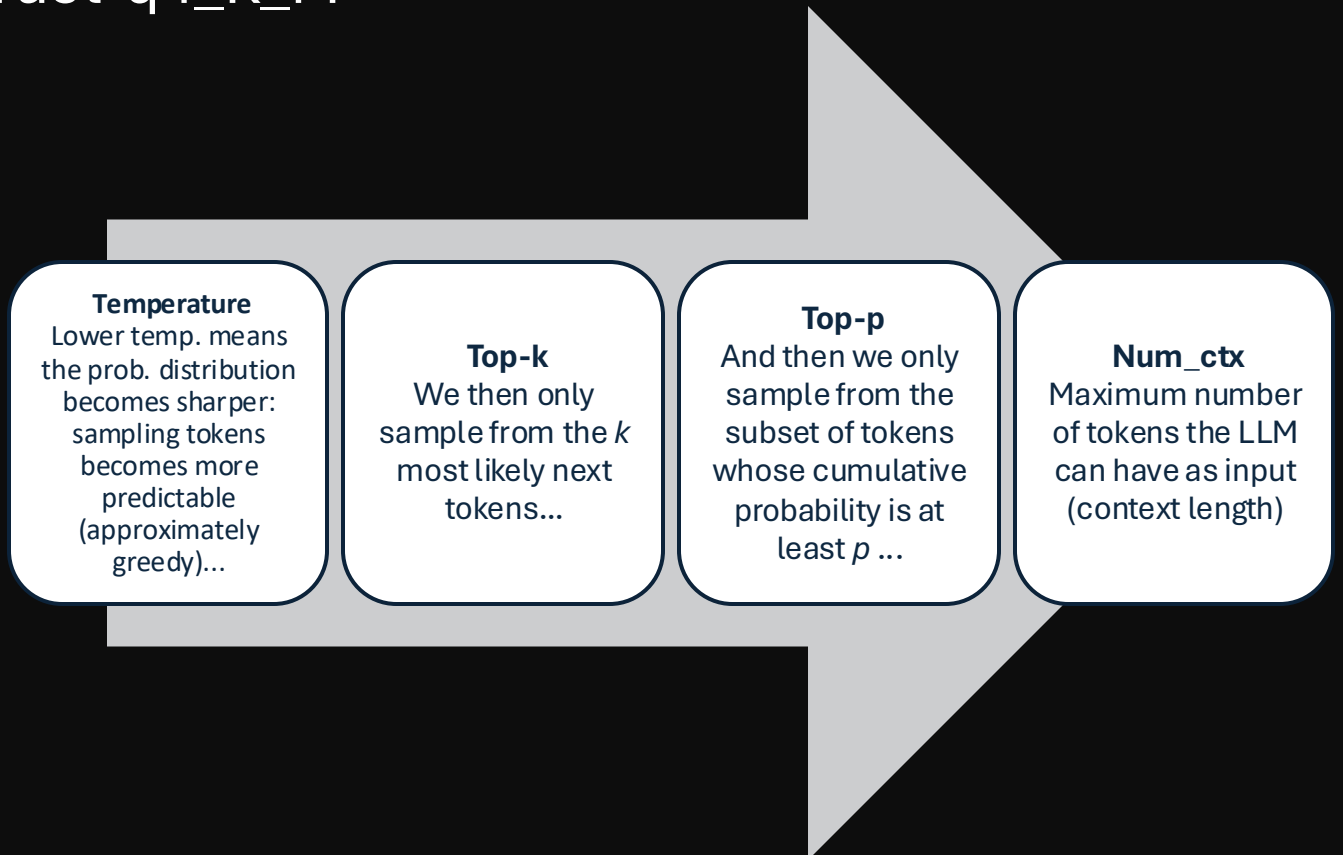
Model properties

Typical example: qwen2.5:7b-instruct-q4_K_M

- Qwen series
- 7 billion parameters (weights)
- Instruct (fine-tuning)
- q4_K_M (make smaller)

Settings:

- **temperature 0.7 >>>**
- **top_k 40 >>>**
- **top_p 0.9 >>>**
- **num_ctx 8192 >>>>**



Local example

```
Windows PowerShell
Total done: 34.1% (of total)
Statements left: 678
Estimated time: 00:02:35:20
Statement: Research is not profit-oriented.
Starting FFP classification for 000707 run 1...
Parsed FFP answer for 000707 run 1: 0
Past run: 3.7 sec
Average run speed: 2.7 s/run
Average statement speed: 13.7 s/statement
Total done: 34.1% (of total)
Statements left: 678
Estimated time: 00:02:35:19
Starting FFP classification for 000707 run 2...
Parsed FFP answer for 000707 run 2: 0
Past run: 2.5 sec
Average run speed: 2.7 s/run
Average statement speed: 13.7 s/statement
Total done: 34.1% (of total)
Statements left: 678
Estimated time: 00:02:35:16
Starting FFP classification for 000707 run 3...
Parsed FFP answer for 000707 run 3: 0
Past run: 2.5 sec
Average run speed: 2.7 s/run
Average statement speed: 13.7 s/statement
Total done: 34.1% (of total)
Statements left: 678
Estimated time: 00:02:35:13
Starting FFP classification for 000707 run 4...
```

Average run speed: 2.7 s/run

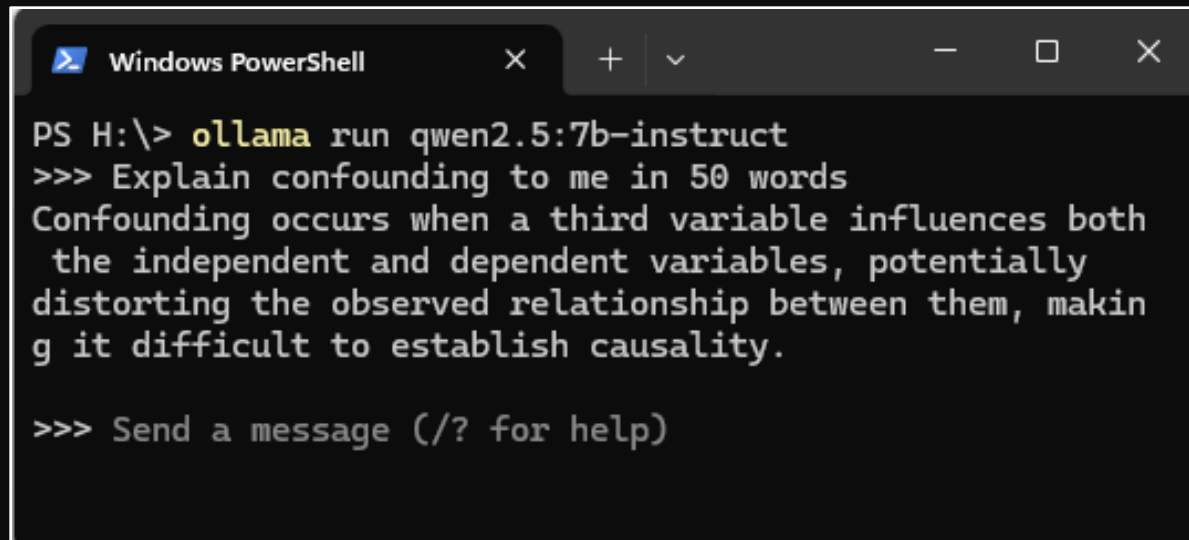
Average statement speed: 13.7 s/statement

Total done: 34.1% (of total)

Statements left: 678

Estimated time: 00:02:35:13

Starting FFP classification for 000707 run 4..



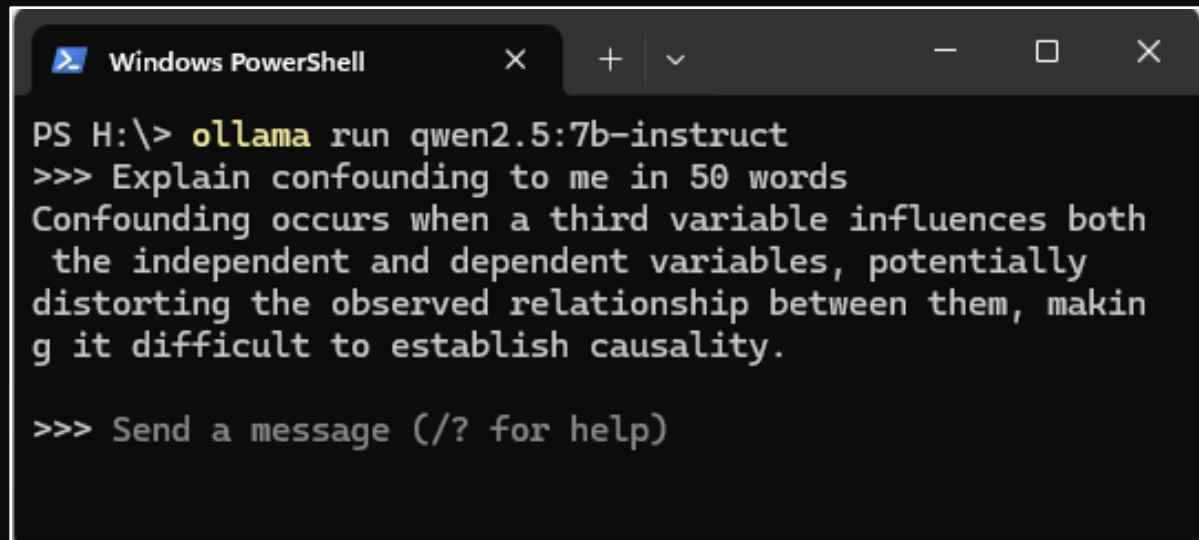
```
Windows PowerShell
PS H:\> ollama run qwen2.5:7b-instruct
>>> Explain confounding to me in 50 words
Confounding occurs when a third variable influences both
the independent and dependent variables, potentially
distorting the observed relationship between them, makin
g it difficult to establish causality.

>>> Send a message (/? for help)
```

Exercise

Use a provider model to do this exercise
(ChatGPT, Copilot, Claude)

- Download a local small model
 - Run it on your laptop
 - Ask it a question
-



```
Windows PowerShell
PS H:\> ollama run qwen2.5:7b-instruct
>>> Explain confounding to me in 50 words
Confounding occurs when a third variable influences both
the independent and dependent variables, potentially
distorting the observed relationship between them, makin
g it difficult to establish causality.

>>> Send a message (/? for help)
```

Exercise

1. Use Ollama to get model: qwen2.5:7b-instruct
 2. Use qwen2.5:7b-instruct: “Explain confounding to me in 50 words”
 3. Do everything in PowerShell (or Terminal on MacBook)
-

Wrapping up

Workflow

Preparation: individual

Preparation: Discussion with your partner

In class: Discussion in the group

In class: Practical approach

After class: Lessons learned

Details will follow

Next sessions:

- WE: 07-10-2026: 1. Literature search
- WE: 21-10-2026: 2. Data collection
- MO: 26-10-2026: 3. Study design
- MO: 02-11-2026: 4. Data analysis
- TH: 12-11-2026: 5. Scientific writing
- MO: 16-11-2026: 6. Peer review
- WE: 25-11-2026: 7. Grant writing

Thank you!

For questions I refer you to Chat / Claude

- Seriously, first try LLMs

For other questions: l.a.pet@lumc.nl

We will send you preparation material for the following course days.

Help us, yourself, and
Digitally Connected



Please scan the code
to indicate presence

